MILESTONE 3: Frontend Development

Quiz Generation - POST API

```
@app.post("/quiz")

async def quiz_api(request: Request):

    data = await request.json()

    text = data.get("text")

    if not text:

        return JSONResponse(content={"error":
"Please provide text for quiz."},
status_code=400)

    quiz = generate_quiz(text)

    print("Generated quiz:", quiz)  #  DEBUG

    return JSONResponse(content={"quiz":
quiz})
```

Learning Recommendations - GET API

```
@app.get("/learn/recommendations")
```

```
async def
learning_recommendation_api(topic: str =
Query(...)):

    recommendation =
get_learning_recommendations(topic)

    return {"topic": topic, "recommendation":
recommendation}
```

Activity 3.1: Build Web Interface

HTML Form (index.html):

Task dropdown (Explain, QnA, Quiz, Summary, Recommend Path)

Text area for user input

Submit button

CSS Styling (style.css):

Responsive design

Styled buttons, input areas, and result container

```
<body>

    <h1>Welcome to EduGenie </h1>
```

```
    <p>Your personal AI tutor for learning support!</p>



    <!-- Q&A -->

    <form id="qaForm">

        <label><strong>Ask EduGenie a Question:</strong></label><br><br>

            <input type="text" id="question" placeholder="Why is the sky blue?" required>

            <button type="submit">Get Answer</button>

    </form>

    <div id="qaResult" class="output"></div>


    <!-- Explanation -->

    <form id="explainForm">

        <label><strong>Need an Explanation?</strong></label><br><br>

            <input type="text" id="topic"
```

```
placeholder="Photosynthesis" required>
        <button type="submit">Explain</button>
    </form>
    <div id="explanationResult"
class="output"></div>
    <!-- Summary -->
    <form id="summaryForm">
        <label><strong>Summarize a
Paragraph:</strong></label><br><br>
        <input type="text" id="summaryText"
placeholder="Paste long content to
summarize" required>
        <button type="submit">Summarize</
button>
    </form>
    <div id="summaryResult" class="output"></
div>
```

```
    <!-- Quiz -->

    <form id="quizForm">

        <label><strong>Generate a Quiz:</
strong></label><br><br>

        <input type="text" id="quizText"
placeholder="Solar System" required>

        <button type="submit">Generate Quiz</
button>

    </form>

    <div id="quizResult" class="output"></div>
```

Fig. Showing Frontend Integration

Activity 3.2: Live Integration

Form submission sends POST request to FastAPI backend.

Result appears below input box in real-time.